



Habib University

CE 321/CS 330 Computer Architecture – Fall 2025

Instructor: Dr. Shehzad Hasan

Time = 40 minutes

Quiz 03

Max Marks: 20

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ID: m110014

Instructions:

- i. Smart watches, laptops, and similar electronics are strictly NOT allowed.
- ii. Answer sheets should contain all steps, working, explanations, and assumptions.
- iii. Attempt the quiz on clean papers with black/blue ink.
- iv. Print your name and HU ID on all sheets.
- v. Turn in your question paper along with your answer sheets.
- vi. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	✓
CLO 4	<i>Analyze the</i> performance of cache operations	

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Question 1 [5 marks]:

Let's say we have a 7-stage pipelined processor. Delay (in ps) for each stage is given below

Fetch	Decode	EXE1	EXE2	EXE3	Memory	Write-back
350	300	400	350	350	450	400

Inter-stage pipeline registers take an additional 40 ps delay

- a) What will be clock cycle time for a single-cycle version of this processor? 2600ps
- b) What will be the clock cycle time for a pipelined version of this processor? 490ps
- c) What is the maximum speed-up we can achieve with this pipelined processor?

$$\frac{2600}{490} = 5.31$$

- d) What is the ideal/theoretical max speed-up we can achieve with a 7-stage pipelined processor, considering no register delays?

$$\frac{2600}{450} = 5.77$$

3.5

- e) Are the speed-ups in part c and d same? If yes, why? If not, why?

No because practical speed up is less than ideal due to register delays & hazard detections. 7 0.5

Question 2 [5 marks]:

Assume the following sequence of instructions is executed on the 5-stage RISC-V pipelined processor

```

LOOP:  ld x2, 64(x23)
        ld x1, 80(x23)
        add x3, x1, x2
        addi x23, x23, -16
        beq x3, x0, LOOP
        add x5, x6, x7
        addi x5, x0, 300
    
```

- a) If there is no forwarding or hazard detection, insert NOPs to ensure correct execution. BTA is calculated in the EX stage. (3 marks)

```

LOOP:  ld x2, 64(x23)
        ld x1, 80(x23)
        add x3, x1, x2
        NOP
        addi x23, x23, -16
        NOP beq x3, x0, LOOP
        NOP add x5, x6, x7
        addi x5, x0, 300
    
```

NOP
NOP
NOP

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- b) If the processor has forwarding in the EX stage, but no hazard detection unit in the ID stage, what happens when the original code executes? (state which instructions will execute incorrectly) (2 marks)

The add x3, x1, x2 instruction will execute incorrectly & due to which the beq x3, x0, loop will also be executed incorrectly. This will also result in the addi x5, x0, 300 being executed incorrectly.

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Question 3 [5 marks]:

Consider the code given below

```

loop: ld  x10, 0(x2)
      add x1, x0, x10
      ld  x11, 4(x1)
      addi x2, x2, 4
      sub x3, x2, x11
      beq x0, x0, loop
      addi x6, x0, 100
    
```

Mux control	Source	Explanation
ForwardA = 00	ID/EX	The first ALU operand comes from the register file.
ForwardA = 10	EX/MEM	The first ALU operand is forwarded from the prior ALU result.
ForwardA = 01	MEM/WB	The first ALU operand is forwarded from data memory or an earlier ALU result.
ForwardB = 00	ID/EX	The second ALU operand comes from the register file.
ForwardB = 10	EX/MEM	The second ALU operand is forwarded from the prior ALU result.
ForwardB = 01	MEM/WB	The second ALU operand is forwarded from data memory or an earlier ALU result.

- a) Assume that the given code runs on a 5-stage pipelined RISC-V processor which has hazard detection unit and forwarding unit for ALU implemented. For this loop, specify values of FwdA, FwdB, PCWrite, IF/ID_Write and Control MUX signals for each cycle in the table below. For a normal instruction, PCWrite and IF/ID_Write both are 1, whereas Control MUX is 0. For a stalled instruction, PCWrite and IF/ID_Write are 0, whereas Control MUX is 1. Values of FwdA and FwdB are given above. All branch decisions are taken in the ID stage. Branches are predicted to be not-taken.

Instr/cycle	1	2	3	4	5	6	7	8	9	10	11	12	13
ld	IF	ID	EX	MEM	WB								
add		IF	ID	EX	MEM	WB							
ld			IF	ID	EX	MEM	WB						
addi			IF	ID	EX	MEM	WB						
sub				IF	ID	EX	MEM	WB					
beq					IF	ID	EX	MEM	WB				
ld						IF	ID	EX	MEM	WB			
add							IF	ID	EX	MEM	WB		
ld								IF	ID	EX	MEM	WB	
addi									IF	ID	EX	MEM	WB
PCWrite	-	1	1	0	0	1	0	0	0	0	0	0	0
IF/ID_Write	-	1	1	0	0	1	0	0	0	0	0	0	0
Control MUX	-	0	0	1	1	0	1	1	1	1	1	1	1
FwdA	-	-	00	00	00	00	00	00	00	00	00	00	00
FwdB	-	-	00	10	01	00	01	00	00	01	01	00	00

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Question 4 [5 marks]:

- a) Write a set of conditions using proper notations for detecting a load-use hazard in the ID stage. (2 marks)

$$\text{if}(\text{ID/EX_rs1} = \text{EX_MEM_rs2} \text{ or } \text{ID/EX_rs2} = \text{EX_MEM_rs1}) \text{ Forward A}$$

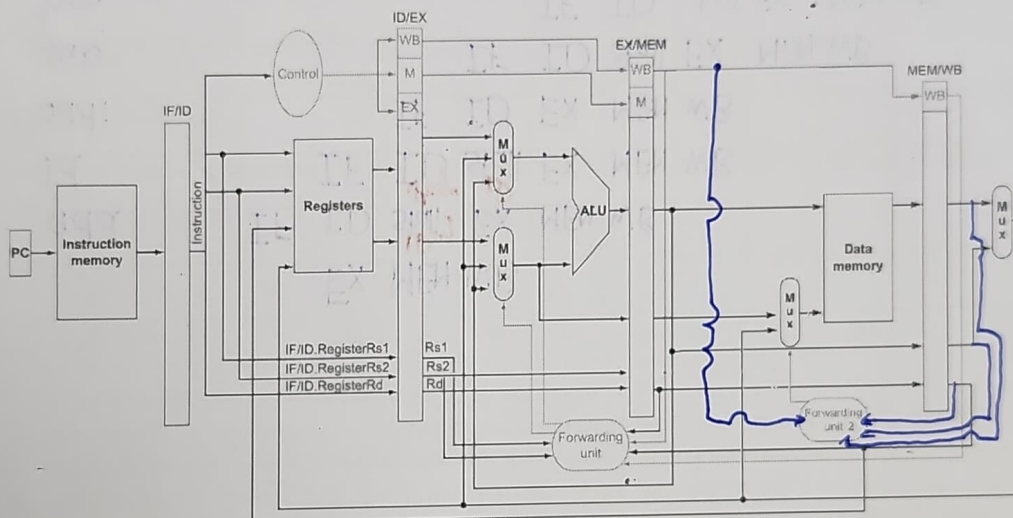
$$\text{if}(\text{ID/EX_rs2} = \text{EX_MEM_rs1} \text{ or } \text{ID/EX_rs1} = \text{EX_MEM_rs2}) \text{ Forward B}$$

- b) A sequence of instructions as shown below can result in data hazards that need stall cycles.

ld x20, 0(x5)

sd x20, 100(x5)

However, this can be resolved using another forwarding unit Forwarding Unit-2 in the MEM stage, which passes on the right value to the memory. This new path would be from the MEM/WB pipeline register back to Data Memory. Write the expression for detecting this sequence and make changes in the diagram by showing all the required connections to Forwarding Unit-2. (3 marks)



Expression

$$\text{or } (\text{EX_MEM_rs2} = \text{MEM_WB_rs1})$$

$$\text{if } (\text{EX_MEM_rs1} = \text{MEM_WB_rs1}) \text{ Forward A}$$

$$\text{if } (\text{EX_MEM_rs2} = \text{MEM_WB_rs2}) \text{ Forward B}$$